

(Marker) Armor stands create bubbles in water

MC-78314 Particles

Status	Resolved / Fixed
Affected	Minecraft 1.8.3, Minecraft 1.8.8, Minecraft 15w36c, Minecraft 15w47c, Minecraft 16w38a, Minecraft 1.11.2 (+96 more)
Fixed in	23w40a
Reported	2015-03-03

STEPS TO REPRODUCE

1. Launch Minecraft in a Creative world using version 21w20a or later.
2. Submerge an area in water, leaving no block directly beneath the position where the armor stand will be summoned.
3. Summon a marker armor stand in the water with `/summon minecraft:armor_stand ~ ~ ~ {Marker:1b}`.
4. Wait approximately 15 seconds (300 ticks).
5. Observe that bubbles appear around the submerged marker armor stand.
6. Place a block directly beneath the armor stand while it remains submerged.
7. Observe that the bubbles stop and the armor stand's air replenishes immediately to 300 seconds.

OBSERVED BEHAVIOUR

Actual behavior

In Java Edition, armor stands use the normal entity air-supply mechanic while submerged in water:

- A newly summoned armor stand starts with `Air:300s` (300 ticks).
- If the stand's eye position is underwater, its air value decreases. After approximately 300 ticks (15 seconds), bubble particles appear.
- Armor stands with `Marker:1b` exhibit this behavior even though their visible model is absent and their hitbox is reduced to a tiny area at their feet.

The following command reproduces the behavior:

```
``mcfuction
/summon minecraft:armor_stand ~ ~ ~ {Marker:1b}
``
```

Submerge the armor stand completely in water and wait approximately 300 ticks. Bubbles appear around the marker armor stand.

There is an apparent interaction with the block directly below a `Marker:1b` armor stand:

- With no block below it, the submerged armor stand's air supply decreases and bubbles eventually appear.
- If any block is placed below it, the air value immediately returns to `300s`, and bubbles stop appearing. This also occurs if the air value had already reached zero or a negative value before the block was placed.
- The block does not need to be a full opaque block. Transparent blocks and partial/non-full blocks, such as glass, slabs, or banners, produce the same result.

For comparison, a normal armor stand (`Marker:0b`) does not produce bubbles when only its feet are submerged. Once its eye position is underwater, however, it also loses air and produces bubbles after approximately 300 ticks. A fully submerged normal armor stand therefore produces bubbles as well.

EXPECTED BEHAVIOUR

When an armor stand is submerged in water, including one with `Marker:1b`, it should remain bubble-free indefinitely rather than producing bubbles after 300 ticks. This behavior should be consistent regardless of whether a block is present beneath the stand, and placing or removing such a block should not alter its air or underwater breathing state.

ENVIRONMENT

Minecraft Java Edition Versions: 1.8.3, 1.8.8, 15w36c, 15w47c, 16w38a, 1.11.2, 17w06a, 1.12, 1.12.1, 1.12.2 Pre-Release 1, 1.12.2 Pre-Release 2, 1.12.2, 17w43a, 17w43b, 17w45b, 17w46a, 17w47a, 17w47b, 17w48a, 18w01a, 18w03b, 18w16a, 18w19b, 18w21a, 18w21b, 18w22a, 1.13-pre1, 1.13-pre2, 1.13-pre3, 1.13-pre4, 1.13-pre6, 1.13-pre7, 1.13-pre8, 1.13-pre9, 1.13-pre10, 1.13, 18w30a, 18w30b, 18w31a, 18w32a, 18w33a, 1.13.1-pre1, 1.13.1, 1.13.2-pre1, 1.13.2-pre2, 1.13.2, 18w47b, 18w48a, 18w48b, 18w49a, 18w50a, 19w02a, 19w03b, 19w03c, 19w04b, 19w05a, 19w06a, 19w07a, 19w12b, 19w13b, 19w14a, 19w14b, 1.15.2 Pre-Release 1, 1.15.2, 20w06a, 20w09a, 20w12a, 20w13a, 20w13b, 20w14a, 20w17a, 20w19a, 1.16 Pre-release 5, 1.16.1, 1.16.2 Pre-release 1, 1.16.2 Release Candidate 1, 1.16.4, 20w46a, 20w48a, 20w49a, 20w51a, 21w03a, 1.16.5, 21w05b, 21w06a, 21w07a, 21w11a, 21w18a, 1.17 Pre-release 3, 1.17, 1.17.1, 1.18.2, 22w13a, 22w14a, 22w18a, 1.19 Pre-release 1, 1.19, 1.19.1, 1.19.2, 1.19.3, 1.19.4 Pre-release 2, 23w33a (includes snapshots)

ORIGINAL DESCRIPTION

Since at least 21w20a, Marker-true-armor stands don't show bubbles if submerged into water as described below, in case there is a block below it. It doesn't matter if it's a full opaque or transparent block, and not only glass, but also e.g. slabs or a banner works.

As soon as there is a block, Air replenishes immediately back to 300s, even if Air went already into negative values and thus bubble appeared prior.

I don't know if this is Wal, I couldn't find anything in the Wiki's armor stand history, but the as-is status could be a potential workaround.

Note to map/contraption makers: If you need the space below a water-submerged, invisible Marker-true-AS to be a nonobstructive void, a barrier block would of course not be an option, but you can use e.g. a structure void block (or e.g. the newish light block). You'll surely come up with better ideas as well.

I don't know if it's Wal that Marker-false-AS still create bubbles if fully submerged in water as described below, and if the above workaround for Marker-true-AS is sufficient for mapmakers, or if they'd prefer a full fix.

I hope Mojang can ask the mapmaking community for their requirements, and, dependent on that, this bugpost may be closed.

As long as this is not officially decided, I'll continue to update this bugpost.

Leaving the old bugpost text below for history and comprehension purposes.

Armor stands with Marker set to true create bubbles underwater which breaks immersion, as those marker-armor stands are only usable/produced in Creative and valuable mapmaker tools. Armor stands get this behaviour as they are part of EntityLiving. As soon as the armor stand's "eye height" is underneath water, bubbles appear after 300 ticks (15 seconds).

That's why an armor stand with Marker false does not create bubbles, as long as it's solely with its "feet" in water, but not with its "head" or rather at its eye height:

As soon as a Marker-false armor stand is underwater at eye height, it'll also create bubbles like the Marker-true armor stand:

As the Marker true armor stand got its tiny hitbox only at its "feet", bubbles appear in any case, when standing in water, after 300 ticks.

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MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-78314>